SoftFuse-KGC: Comment 2 Calibration Experiments

Revision working draft

6 November 2026: precision-corrected evaluation

Completed experiment; standalone insertion draft. This document assembles the new methods and results for review. It is not the integrated manuscript or supplementary submission. Historical metrics must be reconciled before replacing their tables with the standardized results below.

1 Automatic calibration study

1.1 Automatic coefficient calibration and transfer analysis

To investigate dependence on dataset-specific manual coefficients, we distinguish calibration of the original support formula from a complementary calibration baseline with shared feature definitions. The original formula retains its dataset-specific evidence features, while a fixed validation search selects the shortcut penalty, contradiction penalty, and rank-prior mixture from a declared grid of 224 configurations per dataset. Selection uses validation MRR@20 only; the selected configuration is frozen before test evaluation.

The shared-feature baseline uses the original rank prior, source-subgraph contact, bounded log degree, bounded log shared-neighbor count, and a negative direct-contact indicator. These features are computed without reference to evaluation labels and use the same definitions and normalization constants across datasets. They are topological descriptors rather than annotations of therapeutic mechanism. An equal-weight baseline is included to separate effects of the shared features from effects of coefficient learning.

For learned calibration, a linear candidate score is fitted using a pairwise logistic objective with L2 regularization. The primary formulation constrains coefficients to be nonnegative because feature signs are specified explicitly; an unconstrained counterpart is also evaluated. The regularization coefficient is selected by five-fold query-group cross-validation within the existing validation split, which serves as a calibration pool. Candidates known to be other positives for the same calibration-training query are excluded as negative pairs. Queries whose gold entity is absent from the fixed top-20 list cannot contribute a positive training pair, but remain in all evaluation denominators.

We additionally perform leave-one-dataset-out coefficient transfer: the target dataset contributes no validation labels to fitting or hyperparameter selection. Coefficients are fitted on the other datasets, with equal total pair weight per source dataset, and regularization is selected using leave-one-source-dataset-out validation. The target graph, structural retriever, and prepared subgraphs remain available. This experiment therefore assesses transfer of calibration coefficients, not zero-shot completion on an unseen knowledge graph. Pairwise source-to-target transfer, feature ablation, calibration-label learning curves, and coefficient perturbation analyses provide complementary diagnostics.

For fuzzy retrieval, a common edge feature set uses rank-band confidence, candidate incidence, query incidence, direct-contact penalties, and bounded degree information, without dataset-specific relation-family priorities. Automatic selection searches a fixed set of coefficient vectors using validation generated-answer accuracy, with adjusted MRR@20 as a secondary criterion. Candidate order and per-query edge budgets are held fixed. A source-only macro-validation criterion provides the corresponding leave-one-dataset-out edge-coefficient selection. All selected settings are recorded before test inference. The source-only restriction applies to this new coefficient-selection step; the original fuzzy candidate order and per-query budgets are inherited from the historical dataset-specific pipeline. Thus the complete graph-input pipeline is not claimed to be free of prior target-dataset calibration.

End-to-end comparisons use one frozen checkpoint per dataset. Candidate-order interventions retain the full source graph, whereas edge-selection interventions retain the original fuzzy input candidate order. Exact generated-answer accuracy, invalid-output rate, and top-candidate-copy rate are reported separately from adjusted MRR@20. Paired query-cluster bootstrap intervals quantify uncertainty; multiple gold rows for the same query are resampled together. These analyses are retrospective checks on previously studied benchmarks, and their exploratory intervals are not adjusted for multiple comparisons.

Numerical precision correction. Following the nonfinite graph-output audit in the information-retention study, all Comment 2 inference is repeated with graph-encoder checkpoint weights loaded directly in FP32 and

Table 1: Test candidate MRR@20. Original uses the native soft-support formula; the other columns use identical shared feature definitions. LODO excludes target validation labels from coefficient fitting and regularization selection. These comparisons retain exactly the same top-20 candidate membership.

Dataset	Original	Shared equal	Shared learned	Shared LODO
PrimeKG	0.125326	0.125188	0.125326	0.126104
PharmKG	0.028159	0.031883	0.031876	0.031860
Hetionet	0.095439	0.101269	0.115284	0.113579
DRKG	0.046556	0.061426	0.060496	0.058612
repoDB	0.048107	0.057366	0.064902	0.064902

base/PEFT LLM weights loaded directly in BF16. Projected embeddings must remain finite both before and after the BF16 cast. No clipping, sample exclusion, or reuse of legacy FP16 predictions is permitted. Candidate rankings, shared-feature coefficients, and other CPU artifacts remain unchanged. Every validation edge-vector trial is reevaluated; automatic and source-only edge selections are frozen again before test inference, using the same search space, metric definitions, and tie rule.

Identical stable predictions from the Comment 3 run can be reused only when the full checkpoint, model, precision, decoding, and code fingerprint matches, and the exact prompt, candidate ordering, and edge multiset match. This is inference memoization; validation selection does not access test performance. A separate audit compares FP16 encoding, original-checkpoint FP32 encoding, and casts to FP16/BF16 on the actual validation/test graph inputs of both runs. Old–new prediction comparisons identify changed graph inputs after reselection, since these cases cannot isolate precision effects at a fixed input. The precision policy changes both graph and LLM arithmetic, so differences are not attributed causally to either component alone. For the two adaptive graph arms, an additional diagnostic evaluates the old selected coefficients at stable precision. This permits an additive metric decomposition into the precision-policy effect at the fixed old choice and the effect of validation reselection under stable precision. These ten diagnostic dataset–variant combinations are evaluated after the main selections are frozen and do not participate in model selection.

1.2 Results of automatic calibration and coefficient transfer

All five datasets completed the calibration protocol and full-test evaluation. The study includes 1,120 native validation configurations, 20 directed source–target transfers, 100 calibration-label subset runs, and 200 coefficient perturbation runs. Frozen-checkpoint inference evaluates 60 dataset–edge-vector combinations on validation and 55 dataset–variant combinations on test. All inference uses the corrected FP32 graph-encoder/BF16 LLM policy, with validation selections recomputed before test inference. Ten additional test diagnostics hold the previously selected edge coefficients fixed at the corrected precision; they do not participate in configuration selection. The same 2,100 validation rows and 2,100 test rows are reused across their respective variants; these are not independent new examples. Query-level caching avoids repeated generation for identical model inputs.

Candidate-level results show that a common calibration feature space can retain or improve the point estimates of the native soft-support formula. However, the equal-weight control already improves several datasets, so comparisons against the original formula combine feature changes with coefficient learning. Against the feature-matched equal-weight control, learned calibration changes MRR by 0.000138, −0.000007, 0.014015, −0.000930, and 0.007536 on PrimeKG, PharmKG, Hetionet, DRKG, and repoDB, respectively. The exploratory paired query-cluster 95% bootstrap interval excludes zero only for repoDB among these five comparisons: [0.000402, 0.015176]. Thus, the results do not establish that learning uniformly outperforms a common fixed weighting.

The source-only LODO coefficients produce MRR point estimates above the native formula on all five datasets, but they do not uniformly improve over shared equal weights. In particular, DRKG shared equal weights yield 0.061426 versus 0.058612 for LODO. The LODO study concerns coefficient transfer within the existing benchmark pipelines and should not be interpreted as zero-shot completion or generalization to unseen diseases.

Learning curves also differ by dataset. On PharmKG, mean test MRR changes from 0.028975 at 10% of calibration query groups to 0.031856 at 100%, with standard deviations 0.004282 and 0.000020 across subset/CV seeds, respectively. PrimeKG is unchanged at 0.125326 over the tested fractions. Hetionet and repoDB do not show monotonic improvement as the calibration fraction increases. These seed-based standard deviations describe calibration variability, not confidence intervals for performance on independent test sets.

End-to-end outcomes are mixed (Table 2). Higher candidate MRR does not ensure higher generated-answer accuracy. For example, repoDB learned calibration improves candidate MRR but yields no exact answers, compared with four correct answers among 500 rows for shared equal weights. The paired accuracy difference is −0.8 percentage points, with an exploratory 95% interval of [−1.702, −0.190] percentage points. On DRKG, learned calibration yields 2.0% exact accuracy versus 0.8% for equal weights; the corresponding interval for

Table 2: Generated-answer exact accuracy (%) in the candidate-order arm with the full source graph and a fixed checkpoint within each dataset. Accuracy is separate from candidate MRR and adjusted MRR.

Dataset	N	Original	Equal	Learned	LODO
PrimeKG	500	0.0	0.0	0.0	0.0
PharmKG	500	0.0	0.2	0.2	0.2
Hetionet	100	3.0	1.0	1.0	2.0
DRKG	500	1.4	0.8	2.0	1.8
repoDB	500	0.2	0.8	0.0	0.0

Table 3: Test adjusted MRR@20 in the graph-selection arm. Candidate order and the compressed-graph budget are fixed per query. Invalid outputs retain the explicit virtual-rank-one convention described in the protocol; these values must not be read as generated-answer accuracy.

Dataset	Fuzzy	Fixed	Auto	Edge LODO
PrimeKG	0.074687	0.074687	0.074676	0.074676
PharmKG	0.020971	0.020956	0.021317	0.020969
Hetionet	0.100087	0.100087	0.100087	0.100087
DRKG	0.046285	0.046285	0.046285	0.046285
repoDB	0.037776	0.037795	0.038159	0.036653

the 1.2-point difference is [0.000, 2.469] percentage points. These findings limit claims about improvements in generation.

For edge calibration, the numbers of vectors tied at the best validation accuracy and adjusted MRR are 6/12, 2/12, 12/12, 12/12, and 1/12 for PrimeKG, PharmKG, Hetionet, DRKG, and repoDB, respectively. The prespecified tie rule selects the earliest vector, rather than using test results to break ties. The four compressed-graph variants in Table 3 have the same exact-answer accuracy on PrimeKG, PharmKG, Hetionet, and DRKG: 0.0%, 0.2%, 3.0%, and 1.4%, respectively. On repoDB, original fuzzy, shared fixed, and shared auto yield 0.4%, while the newly selected edge-LODO vector yields 0.2%. Thus automatic edge calibration does not establish a uniform exact-answer advantage. Equal metrics under the tested vectors do not identify a unique optimum or prove equivalence for other graphs, budgets, or checkpoints.

Precision correction and validation reselection. The CPU candidate-calibration results remain unchanged. The repeated GPU validation search changes the selected automatic trial on PharmKG (0 to 3) and repoDB (0 to 2), while PrimeKG, Hetionet, and DRKG retain trials 1, 0, and 0. Source-only edge-LODO selections change on all five datasets: 0 to 2, 7 to 2, 7 to 2, 7 to 2, and 1 to 3, respectively. Trial identifiers are zero-based and refer to the unchanged predefined search list. In particular, the old claim that all twelve vectors tie on PharmKG is superseded: only two tie for the best validation score at corrected precision. Even when exact test accuracy is unchanged, individual generated answers and adjusted ranks can change.

Across the 23,100 test row–variant evaluations, 679 normalized generated answers change, and adjusted MRR changes in 15 of the 55 test summaries. These counts include both precision changes and adaptive graph reselection; they are not counts of independent new test examples. Exact accuracy changes in only one summary, repoDB edge-LODO. Holding its old selected coefficients fixed at stable precision retains 0.4% accuracy, whereas the newly selected coefficients yield 0.2%; this difference is therefore attributable to reselection within the stable-precision comparison.

The encoder audit covers 10,765 distinct query/candidate-order/edge-multiset inputs from the legacy and corrected validation/test protocols. It finds 22 nonfinite FP16 calls, all associated with PrimeKG. In each full-source PrimeKG test variant, nine rows have nonfinite FP16 encoder outputs. Casting finite FP32 encoder outputs to FP16 remains nonfinite in six rows of each such variant. All audited FP32 outputs and their BF16 casts are finite. The 22 distinct calls include repeated query cases with different candidate orders; they must not be interpreted as 22 independent affected test examples.

The updated workbook provides every legacy–stable variant comparison and separates changes at fixed graph inputs from changes following reselection. The fixed-old-coefficient diagnostic further decomposes adaptive-arm metric changes into a precision-policy term and a reselection term. Neither term isolates graph arithmetic from LLM arithmetic. Archived FP16 predictions are retained for audit and are not mixed with corrected outputs in the reported tables.

All results above are retrospective analyses of existing benchmark splits. Bootstrap intervals are exploratory and are not adjusted for multiple comparisons. Full configuration traces, coefficients, query-level predictions, feature ablations, transfer matrices, sensitivity analyses, and paired comparisons are provided by the accompanying reproducibility scripts.